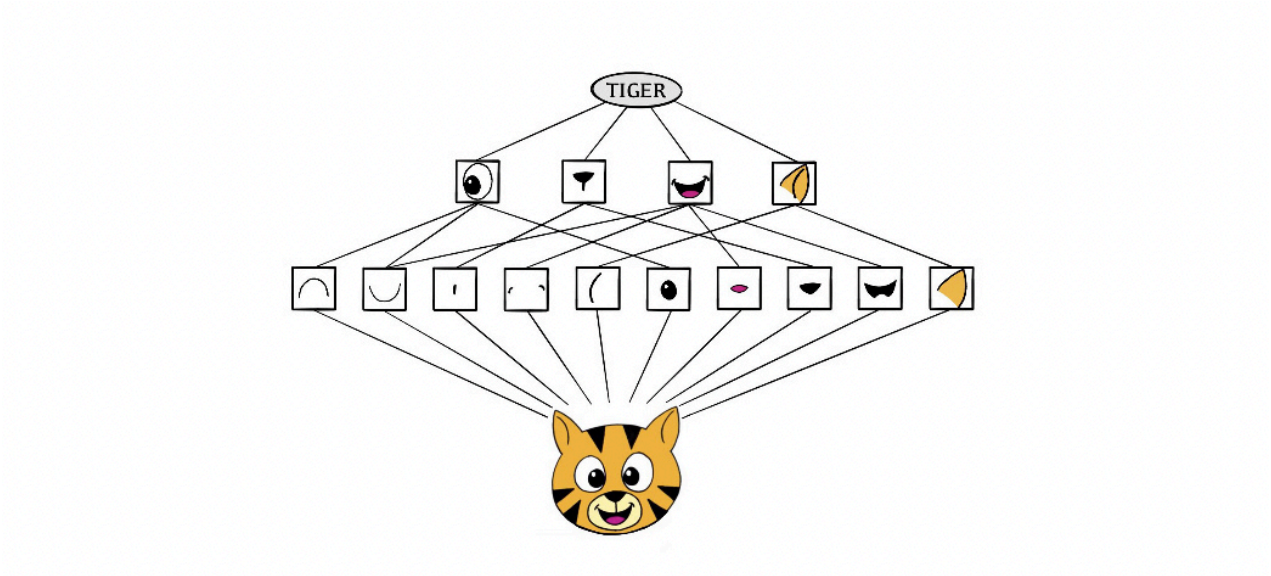
CNNs = Convolutional Neural Networks

Geek4Geek and ISLP

It is popular around 2010.
CNNs also known as **ConvNets** are widely used for Computer vision tasks.

- It mimics to some degree to how human classify images, by recognizing **specific features or patterns anywhere in the images that distinguish each particular object class.**



  The network first identifies low-level feature in the input image, such as small edge, patches of color, and the like. then the low-features are combined to form higher-level features, such as part the ears, mouth and so on. eventually the higher-level feature contribute to the probability of any given output class.

It combines two specialized type of hidden layers, called **convolutional layers** and **pooling layers**.

The convolutional layer search for instances of small patterns in the image, whereas **pooling layers** downsample these to select prominent subset.

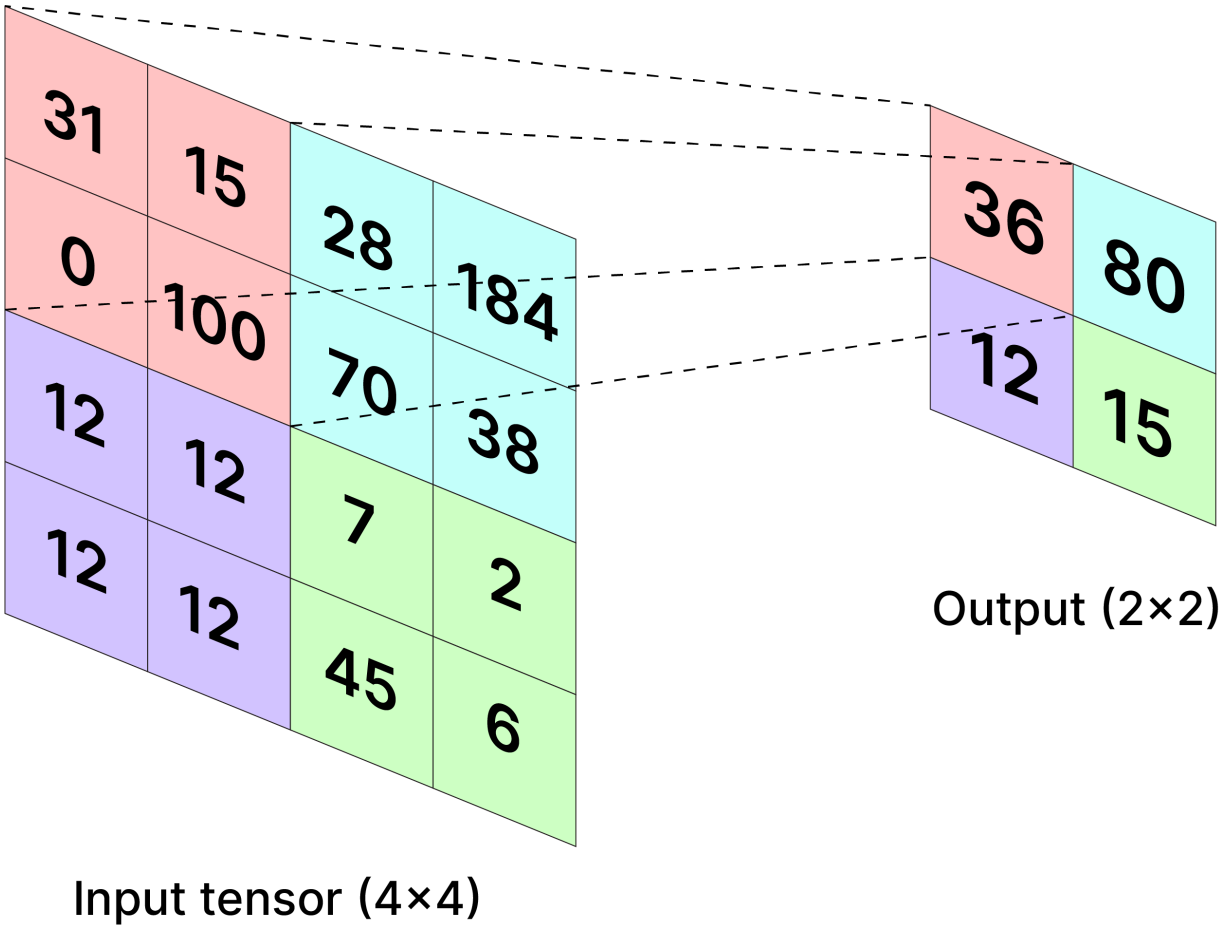
- Automatically learn hierarchical features through convolution operations, from simple edges and textures to complex shapes and objects.
- Detect objects at different positions within an image, ensuring robustness to spatial variations.
- Reduce computational complexity by processing local regions instead of the entire image at once.

Overview of CNNs

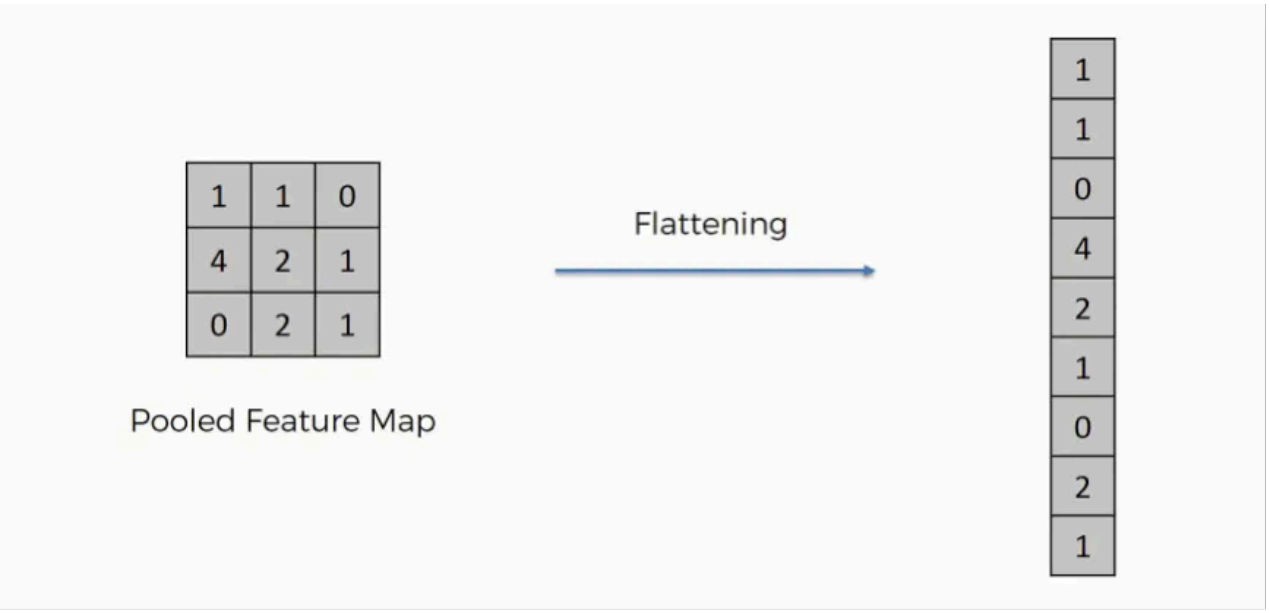
1. **Input Layer**
   It receives the raw image data and passes it to the network for processing. Typically, the input is a 3D volume (Width x height x depth)
   - Stores pixel values of the image (e.g., $32 \times 32 \times 3$ for RGB images).
   - Preserves the spatial structure of the image for further feature extraction.
2. **Convolutional Layer**
   It is responsible for extracting important features from the input data. It applies a set of **learnable filters (kernels)** that slide over the image and compute the dot product between the filter weights and corresponding image patches, producing **feature maps**.
   - Uses small filters (e.g., 2x2, 3x3, 5x5) to scan the input image.
   - Generates feature maps that capture patterns such as edges, textures and shapes.
     Example: Using 12 filters results in an output volume of $32 \times 32 \times 12$.
3. **Activation Layer**
   It introduces non-linearity into the network by applying an **element-wise activation function** to the output of the convolutional layer. This enables the model to learn complex patterns beyond linear relationship.
   - Common activation functions include ReLU, Tanh and Leaky ReLU.
   - Applied element-wise to the feature maps.
   - The output dimensions remain unchanged (e.g., $32 \times 32 \times 12$).
4. **Pooling Layer**
   It condense the **large image** into a **smaller summary image**
   It is used to reduce the spatial dimensions of the feature maps, which help:

   - making computation faster,
   - reducing memory usage and
   - **preventing overfitting**.

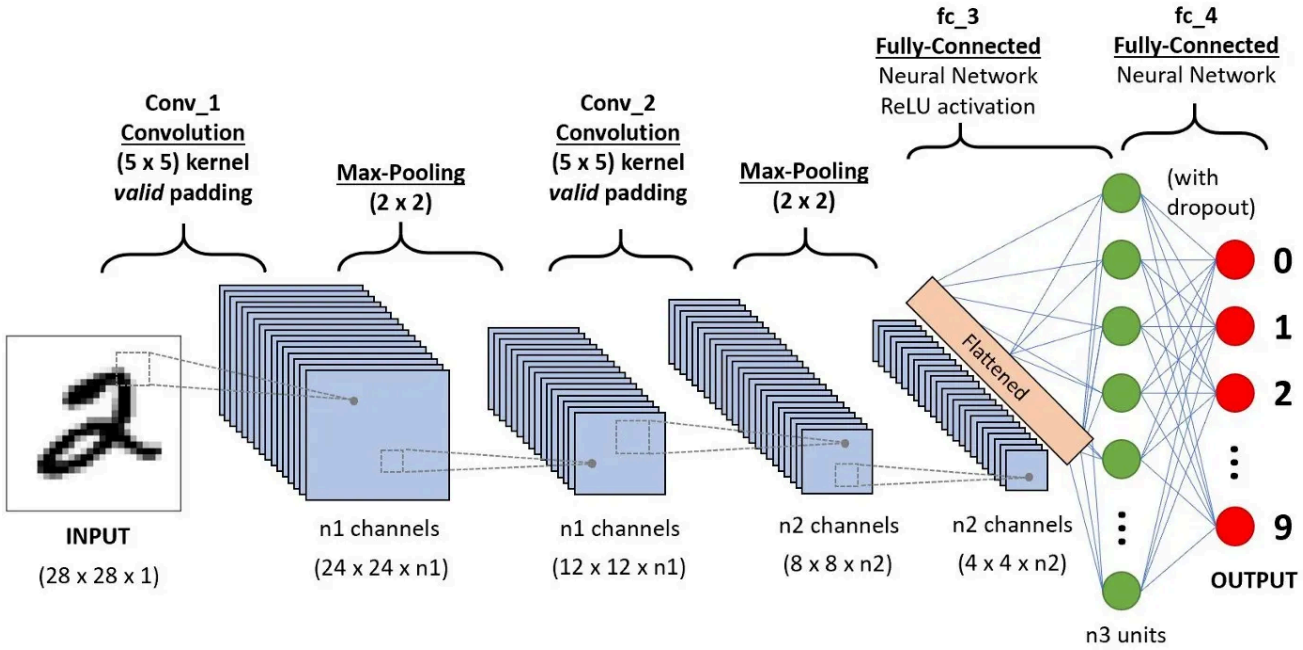
   it is typically inserted between convolutional layers in a CNN.
   - Common types include **Max Pooling** and **Average Pooling**.
   - Reduces **width** and **height** while keeping depth unchanged.
     Example: Using 2 x 2 max pooling with stride 2 reduces the volume from 32 x 32 x 12 to 16 x 16 x 12.



5. **Flattening**
   It converts the **multi-dimensional feature maps** into **a 1-dimensional array for feeding next layer after convolution and pooling**. this vector is then passed to the fully connected layer for classification or regression.
   Example: Flattening 16 x 16 x 12 results in a vector of size 3072 (16 x 16 x 12)



6. **Fully Connected Layer = Connected Components**
   The fully connected (dense) layer performs high-level reasoning using extracted features and produces the final classification scores.
   Example: The 3072-length vector is connected to neurons for classification
7. **Output layer**
   The output layer converts final scores into probabilities using activation function like

Sigmoid (binary classification) or softmax (multi-class classification)

Example: For 10 classes, Softmax produces 10 probability values each representing the likelihood of a class.